

Equal Plücker bodies yield matching generators with an almost-isometric representative

MathlibAnnex.FiniteRecovery.nonempty_almostIsometryMatch_of_all_body_eq

theorem

Selects common minor coordinates with a target almost-isometric representative.

Statement

Let $n = m + 1 > 0$ and $V = \mathbb{R}^n$ with the reference sup norm and standard Lebesgue measure λ . The continuous norms p, q have fixed positive comparison constants

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty, \quad a_q \|x\|_\infty \leq q(x) \leq b_q \|x\|_\infty.$$

Set $K_p = \{x : p(x) \leq 1\}$, $K_q = \{x : q(x) \leq 1\}$ and $v_p = \lambda(K_p)$, $v_q = \lambda(K_q)$, as positive finite real numbers. For a real linear map $A : V \rightarrow \mathbb{R}^N$, let $A_{ij} = (Ae_j)_i$ be its matrix in the standard bases. The minor-coordinate space has one coordinate for each increasing n -tuple I of distinct rows from $\{1, \dots, N\}$, and

$$\Pi_n(A)_I = \det(A_{I,*}).$$

Define the signed generator set and its real convex hull by

$$\mathcal{G}_p^N = \left\{ \sigma v_p \Pi_n(A) : \begin{array}{l} A : V \rightarrow \mathbb{R}^N \text{ is real linear, } \sigma \in \{1, -1\}, \\ \|Ax\|_\infty \leq p(x) \quad (x \in V) \end{array} \right\},$$

$$\mathcal{P}_p^N = \text{conv}_{\mathbb{R}} \mathcal{G}_p^N.$$

Define $\mathcal{G}_q^N, \mathcal{P}_q^N$ in the same way using q, v_q . The generators are vectors of minors, not linear maps. Suppose $\varepsilon > 0$ and $\mathcal{P}_p^N = \mathcal{P}_q^N$ for every N . There exist a dimension N , one minor vector z , and real linear maps $B_p, B_q : V \rightarrow \mathbb{R}^N$ with

$$\forall x, \|B_p x\|_\infty \leq p(x), \quad \forall x, \|B_q x\|_\infty \leq q(x),$$

$$\forall x (q(x) = 1), \quad 1 - \varepsilon < \|B_q x\|_\infty,$$

$$z = \sigma_p v_p \Pi_n(B_p) = \sigma_q v_q \Pi_n(B_q), \quad \sigma_p, \sigma_q \in \{1, -1\}.$$

Assumptions

The two norm models, $\varepsilon > 0$, and body equality for every finite target dimension are inputs. No upper restriction on ε is required.

Conclusion

One dimension and one raw minor vector are shared. Both maps are contractive, but only B_q is required to have the almost lower condition. The signs need not coincide.

Proof route

Choose target satellite parameters, refine the same support on the two compact raw sets, then extract the two representatives.

▼ Proof steps

- 1. Choose parameters for the target norm.** Let $D_q > 0$ be the maximum absolute determinant of q -dominated base rows; its positivity is the determinant-maximum lemma. Put $\eta = D_q/2$. Then $0 < \eta < D_q$. Apply the target satellite package with q, η, ε . It supplies $K > 0$, radius $\min(\varepsilon, 1)/(8K)$, net C_{net} and $w > 0$ with strict satellite budget $< w\eta$. Fix $N = n + |C_{\text{net}}|$ and the oriented linear support ℓ of its selected absolute maximizer.
- 2. Choose a common raw point in this support direction.** For the selected linear functional ℓ , set

$$\mathcal{F}_q = \{z \in \mathcal{G}_q^N : \ell(z) = \max_{u \in \mathcal{G}_q^N} \ell(u)\}.$$

The raw-slice theorem applied to the target parameters says that every $z \in \mathcal{F}_q$ has a real linear representative B_q such that

$$\begin{aligned} \|B_q x\|_\infty &\leq q(x) \quad (x \in V), \\ 1 - \varepsilon &< \|B_q x\|_\infty \quad (q(x) = 1), \quad z = \pm v_q \Pi_n(B_q). \end{aligned}$$

The two raw sets $\mathcal{G}_p^N, \mathcal{G}_q^N$ are compact and nonempty and have equal convex hulls by the input at this N . Apply common raw selection to these sets, this same ℓ , and the displayed property of every point in its target maximum slice. It returns

$$z \in \mathcal{G}_p^N \cap \mathcal{G}_q^N$$

with the displayed target-representative properties. This selects a particular common raw point; it does not infer that the two raw sets are equal.

- 3. Extract two representatives of the same point.** Source raw membership gives B_p , its contractivity and $z = \sigma_p v_p \Pi_n(B_p)$. Target goodness gives B_q , contractivity, lower condition and $z = \sigma_q v_q \Pi_n(B_q)$. Thus

$$\sigma_p v_p \Pi_n(B_p) = z = \sigma_q v_q \Pi_n(B_q).$$

The fixed-dimension construction places these same maps in all nine fields, retaining the separate signs.

The lower condition is required only for the target representative B_q . No lower estimate for B_p or square factorization map is asserted at this stage.

Main citations

- Exact declaration and source
- the determinant-maximum lemma
- the target satellite package
- raw-slice theorem
- compact
- nonempty
- common raw selection
- fixed-dimension construction
- The nine-field matching certificate
- the signed generator set
- its real convex hull

Lean source signature (exact)

```
theorem nonempty_almostIsometryMatch_of_all_body_eq
  {m : ℕ} (MX MY : NormModel (m + 1)) {ε : ℝ} (hε : 0 < ε)
  (hBodies : ∀ N : ℕ, MathlibAnnex.PluckerBody.body MX N = MathlibAnnex.PluckerBody.body
MY N) :
  Nonempty (AlmostIsometryMatch MX MY ε)
```

In the source	Mathematical meaning
<code>Fin (m + 1) → ℝ; →L[ℝ]</code>	The reference $V = \mathbb{R}^n$, $n = m + 1 > 0$, with sup norm; arrows mean continuous real linear maps. Source indices $0, \dots, n - 1$ correspond to formula indices $1, \dots, n$.
<code>MX MY : NormModel (m + 1); MX.p; MY.p</code>	<code>NormModel n</code> is the source alias for <code>EquivalentSeminorm (Fin n → ℝ)</code> . Thus these are the two continuous norms p, q , with $a_p, b_p, a_q, b_q > 0$ and the displayed two-sided comparisons. The applications <code>MX.p x</code> and <code>MY.p x</code> mean $p(x), q(x)$.
<code>MX.closedUnitBall; MY.closedUnitBall; MX.closedUnitBallVolume; MY.closedUnitBallVolume</code>	The sets K_p, K_q and their real Lebesgue volumes v_p, v_q . Real volumes are the <code>toReal</code> of the finite extended measures.
<code>hBodies : ∀ N : ℕ, PluckerBody.body MX N = PluckerBody.body MY N</code>	For every nonnegative N , the real convex hulls $\mathcal{P}_p^N$ and $\mathcal{P}_q^N$ are equal. The source body is the convex hull of the vectors $\pm v_p \Pi_n(A)$, where $A : V \rightarrow \mathbb{R}^N$ is real linear and $\ Ax\ _\infty \leq p(x)$ for every x . The target body uses q, v_q in place of p, v_p .
<code>hε : 0 < ε; Nonempty (AlmostIsometryMatch MX MY ε)</code>	Positive ε and existence of one record carrying all nine fields below. <code>Nonempty</code> asserts existence, not uniqueness.

The inputs before the final colon are the two norm models, the positive-error hypothesis and all body equalities. The conclusion `Nonempty (...)` asserts existence of a record with the fields displayed separately below.

Related structure, shown separately: The nine-field matching certificate.

```
structure AlmostIsometryMatch {m : ℕ}
  (MX MY : NormModel (m + 1)) (ε : ℝ) where
  ambientDim : ℕ
  z : PluckerCoord (m + 1) ambientDim
  sourceMap : Coord (m + 1) →L[ℝ] SupCoord ambientDim
  targetMap : Coord (m + 1) →L[ℝ] SupCoord ambientDim
  isContraction_sourceMap : MX.IsContraction sourceMap
  isContraction_targetMap : MY.IsContraction targetMap
  finMapAlmostIsometric_targetMap : FinMapAlmostIsometric MY ε targetMap
  eq_ballVolumeScaledMaximalMinors_sourceMap_or_neg :
    z = MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors MX sourceMap V
    z = - MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors MX sourceMap
  eq_ballVolumeScaledMaximalMinors_targetMap_or_neg :
    z = MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors MY targetMap V
    z = - MathlibAnnex.Matrix.ballVolumeScaledMaximalMinors MY targetMap
```

In the source	Mathematical meaning
ambientDim : ℕ	The selected nonnegative integer N .
z : PluckerCoord (m + 1) ambientDim	The common vector z , whose coordinates are indexed by increasing n -tuples of rows.
sourceMap : Coord (m + 1) →L[ℝ] SupCoord ambientDim	The continuous real linear map $B_p : V \rightarrow \mathbb{R}^N$.
targetMap : Coord (m + 1) →L[ℝ] SupCoord ambientDim	The continuous real linear map $B_q : V \rightarrow \mathbb{R}^N$.
isContraction_sourceMap	The assertion $\ B_p x\ _\infty \leq p(x)$ for every $x \in V$.
isContraction_targetMap	The assertion $\ B_q x\ _\infty \leq q(x)$ for every $x \in V$.
finMapAlmostIsometric_targetMap	For every x with $q(x) = 1$, $1 - \varepsilon < \ B_q x\ _\infty$. This is only a property of the target map.
In the source	Mathematical meaning
eq_ballVolumeScaledMaximalMinors_sourceMap_or_neg	$z = v_p \Pi_n(B_p)$ or $z = -v_p \Pi_n(B_p)$. The disjunction is represented by the one sign $\sigma_p \in \{1, -1\}$.
eq_ballVolumeScaledMaximalMinors_targetMap_or_neg	$z = v_q \Pi_n(B_q)$ or $z = -v_q \Pi_n(B_q)$. It concerns the same z , but its sign σ_q need not equal σ_p .

The clauses concern one common record; $\forall$ means for every and $\vee$ means or.

Exact definitions: the signed generator set; its real convex hull.

Surrounding assumptions and aliases.

▼ **Exact content identity**

Declaration: `MathlibAnnex.FiniteRecovery.nonempty_almostIsometryMatch_of_all_body_eq`

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